NDIC - DATEX II Measured Data Publication - Detector Data

Release 1.0rev1

Národní Dopravní Informační Centrum

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This Specification defines data format, used for publishing traffic data (flow, occupancy and speed) for set of predefined traffic detector sites.

CHAPTER

ONE

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format
: description of data structure for transferring some kind of information

protocol
: description of data transfer process and rules

NDIC
: national traffic information centre

DATEX II model
: class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [d2ref]. For additional information see [d2supp].

(DATEX II) profile
: simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension
extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension
Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension
Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication
an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema
language, which allows definition of rules for a XML document structure.

XSD
XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri
uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CHAPTER TWO

CONCEPTS

Described format allows publishing traffic data (flow, occupancy and speed) for set of predefined traffic detector sites.

2.1 Publication type MeasuredDataPublication

Format is based on DATEX II publication of type *MeasuredDataPublication* [16157-5] and is used to publish traffic data for a set of predefined measuring equipment.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="MeasuredDataPublication">
    <feedType>x-format:cz-ndic_d2-detector-data-v1.0</feedType>
    <publicationTime>2018-03-29T08:00:29+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </publicationCreator>
```

Traffic data is provided in element *measurementSiteReference* for a particular measuring equipment defined in a static publication *MeasurementSiteTablePublication* that is referenced via its unique id and version in the element *measurementSiteTableReference*.

Element *measurementSiteReference* with a set of measurements for a particular measuring site should appear in *siteMeasurements* multiple times (each time for different detector).

2.2 Predefined measurement sites

The *measuring equipment* capability and location does not change. The measurement station is therefore, in this profile, described by a reference (id) to a predefined measurement site table publication defined in different document. The reference points to a class instance *MeasurementSiteRecord* from package *MeasurementSiteTablePublication*. Therefore, the location and capabilities of the equipment is not described directly in the file.

The equipment information (location and capabilities) is determined by reference to a set of predefined equipment. To look up given equipment, it is necessary to obtain publication *MeasurementSiteTablePublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *MeasurementSiteRecord*), with the same identifier and version located within the same measurement table.

Fully qualified identifier of the measurement site consists of:

- measurement site table id and version and by
- measurement site record id and version.

```
    <measurementSiteTableReference id="CZ_DRU_MST_DET" targetClass=
↪"MeasurementSiteTable" version="1"/>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <siteMeasurements>
      <measurementSiteReference id="1307" targetClass="MeasurementSiteRecord" version=
↪"1"/>
      <measurementTimeDefault>2018-03-29T08:00:29+02:00</measurementTimeDefault>
```

2.3 Traffic detector capabilities

Each detector, in this profile, is capable of different set of measurements:

- traffic concentration,
- traffic flow,
- traffic speed.

The measurements, can be, in the *MeasurementSiteTablePublication*, further specified by positional arguments and vehicle specification (i.e. traffic flow of heavy vehicles, road temperature in second lane).

The full list of possible measurements given by the detector is called a capability and set in the *MeasurementSiteTablePublication* for each detector verbatim as a number of *measurementSpecificCharacteristics* identified within the record by an index. To refer to that particular capability (i.e. traffic flow for lorries) from this publication the capability index is used.

```
      <measuredValue index="1">
        <measuredValue>
          <basicData xsi:type="TrafficFlow">
            <vehicleFlow>
              <vehicleFlowRate>29</vehicleFlowRate>
            </vehicleFlow>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="2">
        <measuredValue>
          <basicData xsi:type="TrafficConcentration">
```

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```
            <occupancy>
              <percentage>0</percentage>
            </occupancy>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="3">
        <measuredValue>
          <basicData xsi:type="TrafficSpeed">
            <averageVehicleSpeed>
              <speed>120</speed>
            </averageVehicleSpeed>
          </basicData>
        </measuredValue>
      </measuredValue>
```

Traffic flow is represented by vehicles per hour, occupancy as a percentage and speed in km per hour. Capability specification, i.e. its relation to a vehicle category or lane location is not expressed in dynamic publication.

2.4 Publishing only currently available data

Format provides information about capabilities with known status. Either value is known (could be zero) or the capability is in faulty state (i.e. unknown). Only situation when respective capability *measuredValue* with given index value is not present is when the value could not be calculated (i.e. for speeds per vehicle category, when that specific category was no present in a vehicle flow).

CHAPTER

THREE

TIME VALIDITY

The time of validity for all measured data at one measurement site is given by the mandatory element *measuremetnTimeDefault* nested in *measurementSiteReference*. So each measurement site has to have its own validity time set up. Validity time means end of the measurement period, when data is calculated and provided.

No further specification of validity is used.

```
    <feedType>x-format:cz-ndic_d2-detector-data-v1.0</feedType>
    <publicationTime>2018-03-29T08:00:29+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </publicationCreator>
    <measurementSiteTableReference id="CZ_DRU_MST_DET" targetClass="MeasurementSiteTable" version="1"/>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <siteMeasurements>
      <measurementSiteReference id="1307" targetClass="MeasurementSiteRecord" version="1"/>
      <measurementTimeDefault>2018-03-29T08:00:29+02:00</measurementTimeDefault>
```

CHAPTER

FOUR

LOCATION REFERENCING

The format does not describe locations, they are expected as a part of equipment specification in publication *MeasurementSiteTablePublication* [16157-5].

Location referencing methods are out of scope of this document and are described elsewhere.

CHAPTER

FIVE

APPENDIX A: MESSAGE EXAMPLES

5.1 detector-data.xml - traffic data measured at one equipment

Publication contains one record for one equipment capable to measure several characteristics.

Equipment details (capabilities and location) are defined in different publication of type *MeasurementSiteTablePublication*, which is not described here in detail.

The equipment, is identified by a measurement site table reference *id=”CZ_DRU_MST_DET”* (with version), and by measurement site reference *id=”1307”*, (with version).

```xml
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsd=
↪"http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
↪instance">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="MeasuredDataPublication">
    <feedType>x-format:cz-ndic_d2-detector-data-v1.0</feedType>
    <publicationTime>2018-03-29T08:00:29+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>DRU</nationalIdentifier>
    </publicationCreator>
    <measurementSiteTableReference id="CZ_DRU_MST_DET" targetClass=
↪"MeasurementSiteTable" version="1"/>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <siteMeasurements>
      <measurementSiteReference id="1307" targetClass="MeasurementSiteRecord" version=
↪"1"/>
      <measurementTimeDefault>2018-03-29T08:00:29+02:00</measurementTimeDefault>
      <!--start of overall characteristics: flow, occupancy and speed -->
      <measuredValue index="1">
        <measuredValue>
          <basicData xsi:type="TrafficFlow">
            <vehicleFlow>
              <vehicleFlowRate>29</vehicleFlowRate>
            </vehicleFlow>
          </basicData>
```

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```
    </measuredValue>
  </measuredValue>
  <measuredValue index="2">
    <measuredValue>
      <basicData xsi:type="TrafficConcentration">
        <occupancy>
          <percentage>0</percentage>
        </occupancy>
      </basicData>
    </measuredValue>
  </measuredValue>
  <measuredValue index="3">
    <measuredValue>
      <basicData xsi:type="TrafficSpeed">
        <averageVehicleSpeed>
          <speed>120</speed>
        </averageVehicleSpeed>
      </basicData>
    </measuredValue>
  </measuredValue>
  <!--end of overall characteristics: flow, occupancy and speed -->
  <!--start of vehicle flow per hour, classes: 17-21 -->
  <measuredValue index="4">
    <measuredValue>
      <basicData xsi:type="TrafficFlow">
        <vehicleFlow>
          <vehicleFlowRate>23</vehicleFlowRate>
        </vehicleFlow>
      </basicData>
    </measuredValue>
  </measuredValue>
  <measuredValue index="5">
    <measuredValue>
      <basicData xsi:type="TrafficFlow">
        <vehicleFlow>
          <vehicleFlowRate>0</vehicleFlowRate>
        </vehicleFlow>
      </basicData>
    </measuredValue>
  </measuredValue>
  <measuredValue index="6">
    <measuredValue>
      <basicData xsi:type="TrafficFlow">
        <vehicleFlow>
          <vehicleFlowRate>4</vehicleFlowRate>
        </vehicleFlow>
      </basicData>
    </measuredValue>
  </measuredValue>
  <measuredValue index="7">
    <measuredValue>
      <basicData xsi:type="TrafficFlow">
        <vehicleFlow>
          <vehicleFlowRate>2</vehicleFlowRate>
        </vehicleFlow>
      </basicData>
```

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```
        </measuredValue>
      </measuredValue>
      <measuredValue index="8">
        <measuredValue>
          <basicData xsi:type="TrafficFlow">
            <vehicleFlow>
              <vehicleFlowRate>0</vehicleFlowRate>
            </vehicleFlow>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="9">
        <measuredValue>
          <basicData xsi:type="TrafficFlow">
            <vehicleFlow>
              <vehicleFlowRate>0</vehicleFlowRate>
            </vehicleFlow>
          </basicData>
        </measuredValue>
      </measuredValue>
      <!--end of vehicle flow per hour, classes: 17-21 -->
      <!--start of occupancy, classes: 17-21 with one error value -->
      <measuredValue index="10">
       <measuredValue>
          <measurementEquipmentFault>
            <faultLastUpdateTime>2018-03-21T15:59:21+02:00</faultLastUpdateTime>
            <measurementEquipmentFault>noDataValuesAvailable</
↪measurementEquipmentFault>
          </measurementEquipmentFault>
        </measuredValue>
      </measuredValue>
      <measuredValue index="11">
        <measuredValue>
          <basicData xsi:type="TrafficConcentration">
            <occupancy>
              <percentage>0</percentage>
            </occupancy>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="12">
        <measuredValue>
          <basicData xsi:type="TrafficConcentration">
            <occupancy>
              <percentage>0</percentage>
            </occupancy>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="13">
        <measuredValue>
          <basicData xsi:type="TrafficConcentration">
            <occupancy>
              <percentage>0</percentage>
            </occupancy>
          </basicData>
```

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```
        </measuredValue>
      </measuredValue>
      <measuredValue index="14">
        <measuredValue>
          <basicData xsi:type="TrafficConcentration">
            <occupancy>
              <percentage>0</percentage>
            </occupancy>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="15">
        <measuredValue>
          <basicData xsi:type="TrafficConcentration">
            <occupancy>
              <percentage>0</percentage>
            </occupancy>
          </basicData>
        </measuredValue>
      </measuredValue>
      <!--end of occupancy, classes: 17-21 -->
      <!--start averagge speed, classes: 16,18,19 -->
      <measuredValue index="16">
        <measuredValue>
          <basicData xsi:type="TrafficSpeed">
            <averageVehicleSpeed>
              <speed>129</speed>
            </averageVehicleSpeed>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="18">
        <measuredValue>
          <basicData xsi:type="TrafficSpeed">
            <averageVehicleSpeed>
              <speed>87</speed>
            </averageVehicleSpeed>
          </basicData>
        </measuredValue>
      </measuredValue>
      <measuredValue index="19">
        <measuredValue>
          <basicData xsi:type="TrafficSpeed">
            <averageVehicleSpeed>
              <speed>89</speed>
            </averageVehicleSpeed>
          </basicData>
        </measuredValue>
      </measuredValue>
      <!--end averagge speed, classes: 16,18,19 -->
    </siteMeasurements>
  </payloadPublication>
</d2LogicalModel>
```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information, measurement site table reference *measurementSiteTableReference* and one element *siteMeasurements*.

Attribute *xsi:type="MeasuredDataPublication"* in element *payloadPublication* determines, that this envelope is of type Measured data publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* gives publication time for the whole publication which might be the time, for which were all traffic status values calculated (start of validity).

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of traffic status at one place

The element *siteMeasurements* contains one or more measurements referenced to a particular road equipment *measurementSiteReference*. the *measurementSiteReference* is identified by id and version.

measurementTimeDefault element gives time of validity for the set of measurement within one measurement site. It is also the time, when measured traffic data were created (i.e. the end of the measurement period).

Following element *measuredValue* can be repeated and describes a measured traffic data for a particular measurement site.

The equipment (traffic detector) has 21 capabilities defined. They are vehicle categorized versions of three traffic variables:

- traffic flow,
- traffic concentration,
- traffic speed.

First 3 capabilities referenced in *measuredValue* with indexes 1-3 describe flow, occupancy and speed for all vehicles. Remaining capabilities describe these traffic values categorized by vehicle type.

E.g. capabilities with indexes 4-9 describe traffic flow for:

- car,
- lorry,
- trailer,
- van,
- bus,
- motorcycle.

The capabilities *measuredValue* with indexes 10-15 describe categorized occupancy and 16-21 categorized speed.

Elements *measuredValue* with indexes 17, 20 and 21 are not present since the speed value could not be calculated (no vehicles of that class within measurement period).

Nested element *basicData* contains the actual measured value of the given type.

Time validity

Start of validity for measured data is stated by mandatory element *measurementTimeDefault*.

CHAPTER

SIX

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```
<?xml version="1.0" encoding="utf-8" standalone="no"?>
<xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified" xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace="http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="_ExtensionType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="_MeasurementSiteRecordVersionedReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:VersionedReference">
        <xs:attribute name="targetClass" use="required" fixed="MeasurementSiteRecord" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="_MeasurementSiteTableVersionedReference">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:VersionedReference">
        <xs:attribute name="targetClass" use="required" fixed="MeasurementSiteTable" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="_SiteMeasurementsIndexMeasuredValue">
    <xs:sequence>
      <xs:element name="measuredValue" type="D2LogicalModel:MeasuredValue" minOccurs="1" maxOccurs="1" />
    </xs:sequence>
    <xs:attribute name="index" type="xs:int" use="required" />
  </xs:complexType>
  <xs:complexType name="_VehicleCharacteristicsExtensionType">
    <xs:sequence>
      <xs:element name="vehicleCharacteristicsExtended" type="D2LogicalModel:VehicleCharacteristicsExtended" minOccurs="0" />
      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
```

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```
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="AreaOfInterestEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="continentWide" />
      <xs:enumeration value="national" />
      <xs:enumeration value="neighbouringCountries" />
      <xs:enumeration value="notSpecified" />
      <xs:enumeration value="regional" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="AxleFlowValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour"␣
↪minOccurs="1" maxOccurs="1" />
          <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="AxlesPerHour">
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="BasicData" abstract="true">
    <xs:sequence>
      <xs:element name="measurementOrCalculationPeriod" type="D2LogicalModel:Seconds"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="measurementOrCalculationTime" type="D2LogicalModel:DateTime"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="basicDataExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="measurementOrCalculatedTimePrecision" type=
↪"D2LogicalModel:TimePrecisionEnum" use="optional" />
  </xs:complexType>
  <xs:simpleType name="Boolean">
    <xs:restriction base="xs:boolean" />
  </xs:simpleType>
  <xs:simpleType name="ComparisonOperatorEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="equalTo" />
      <xs:enumeration value="greaterThan" />
      <xs:enumeration value="greaterThanOrEqualTo" />
      <xs:enumeration value="lessThan" />
      <xs:enumeration value="lessThanOrEqualTo" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="ComputationMethodEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
↪>
      <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
      <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
```

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```xml
      <xs:enumeration value="medianOfSamplesInATimePeriod" />
      <xs:enumeration value="movingAverageOfSamples" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="ConcentrationOfVehiclesValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="concentrationOfVehicles" type="D2LogicalModel:ConcentrationVehiclesPerKilometre" minOccurs="1" maxOccurs="1" />
          <xs:element name="concentrationOfVehiclesValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="ConcentrationVehiclesPerKilometre">
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:simpleType name="ConfidentialityValueEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="internalUse" />
      <xs:enumeration value="noRestriction" />
      <xs:enumeration value="restrictedToAuthorities" />
      <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
      <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="CountryEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="at" />
      <xs:enumeration value="be" />
      <xs:enumeration value="bg" />
      <xs:enumeration value="ch" />
      <xs:enumeration value="cs" />
      <xs:enumeration value="cy" />
      <xs:enumeration value="cz" />
      <xs:enumeration value="de" />
      <xs:enumeration value="dk" />
      <xs:enumeration value="ee" />
      <xs:enumeration value="es" />
      <xs:enumeration value="fi" />
      <xs:enumeration value="fo" />
      <xs:enumeration value="fr" />
      <xs:enumeration value="gb" />
      <xs:enumeration value="gg" />
      <xs:enumeration value="gi" />
      <xs:enumeration value="gr" />
      <xs:enumeration value="hr" />
      <xs:enumeration value="hu" />
      <xs:enumeration value="ie" />
      <xs:enumeration value="im" />
      <xs:enumeration value="is" />
      <xs:enumeration value="it" />
      <xs:enumeration value="je" />
```

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```
      <xs:enumeration value="li" />
      <xs:enumeration value="lt" />
      <xs:enumeration value="lu" />
      <xs:enumeration value="lv" />
      <xs:enumeration value="ma" />
      <xs:enumeration value="mc" />
      <xs:enumeration value="mk" />
      <xs:enumeration value="mt" />
      <xs:enumeration value="nl" />
      <xs:enumeration value="no" />
      <xs:enumeration value="pl" />
      <xs:enumeration value="pt" />
      <xs:enumeration value="ro" />
      <xs:enumeration value="se" />
      <xs:enumeration value="si" />
      <xs:enumeration value="sk" />
      <xs:enumeration value="sm" />
      <xs:enumeration value="tr" />
      <xs:enumeration value="va" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel" />
  <xs:complexType name="D2LogicalModel">
    <xs:sequence>
      <xs:element name="exchange" type="D2LogicalModel:Exchange" />
      <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication" minOccurs="0" />
      <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
  </xs:complexType>
  <xs:complexType name="DataValue" abstract="true">
    <xs:sequence>
      <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0" maxOccurs="1" />
      <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
      <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
    <xs:attribute name="computationalMethod" type="D2LogicalModel:ComputationMethodEnum" use="optional" />
    <xs:attribute name="numberOfIncompleteInputs" type="D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="numberOfInputValuesUsed" type="D2LogicalModel:NonNegativeInteger" use="optional" />
    <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
    <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional" />
    <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage" use="optional" />
  </xs:complexType>
  <xs:simpleType name="DateTime">
```

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```xml
    <xs:restriction base="xs:dateTime" />
  </xs:simpleType>
  <xs:complexType name="DateTimeValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1" maxOccurs="1" />
          <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="DurationValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="duration" type="D2LogicalModel:Seconds" minOccurs="1" maxOccurs="1" />
          <xs:element name="durationValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="ElaboratedDataFault">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Fault">
        <xs:sequence>
          <xs:element name="elaboratedDataFault" type="D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
          <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="ElaboratedDataFaultEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="intermittentDataValues" />
      <xs:enumeration value="noDataValuesAvailable" />
      <xs:enumeration value="spuriousUnreliableDataValues" />
      <xs:enumeration value="unspecifiedOrUnknownFault" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="Exchange">
    <xs:sequence>
      <xs:element name="supplierIdentification" type="D2LogicalModel:InternationalIdentifier" />
      <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="Fault">
```

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```xml
    <xs:sequence>
      <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
      <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
      <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0" maxOccurs="1" />
      <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs="1" maxOccurs="1" />
      <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum" minOccurs="0" maxOccurs="1" />
      <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="FaultSeverityEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="low" />
      <xs:enumeration value="medium" />
      <xs:enumeration value="high" />
      <xs:enumeration value="unknown" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="Float">
    <xs:restriction base="xs:float" />
  </xs:simpleType>
  <xs:complexType name="FloatingPointMetreDistanceValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="floatingPointMetreDistance" type="D2LogicalModel:MetresAsFloat" minOccurs="1" maxOccurs="1" />
          <xs:element name="floatingPointMetreDistanceValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="FuelType2Enum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="all" />
      <xs:enumeration value="petrol95Octane" />
      <xs:enumeration value="petrol98Octane" />
      <xs:enumeration value="petrolLeaded" />
      <xs:enumeration value="petrolUnleaded" />
      <xs:enumeration value="unknown" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="FuelTypeEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="battery" />
      <xs:enumeration value="biodiesel" />
      <xs:enumeration value="diesel" />
      <xs:enumeration value="dieselBatteryHybrid" />
      <xs:enumeration value="ethanol" />
```

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```
      <xs:enumeration value="hydrogen" />
      <xs:enumeration value="liquidGas" />
      <xs:enumeration value="lpg" />
      <xs:enumeration value="methane" />
      <xs:enumeration value="petrol" />
      <xs:enumeration value="petrolBatteryHybrid" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="GrossWeightCharacteristic">
    <xs:sequence>
      <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="grossVehicleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
↪" maxOccurs="1" />
      <xs:element name="grossWeightCharacteristicExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HeaderInformation">
    <xs:sequence>
      <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"␣
↪minOccurs="0" maxOccurs="1" />
      <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
↪" minOccurs="1" maxOccurs="1" />
      <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
↪" minOccurs="1" maxOccurs="1" />
      <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HeaviestAxleWeightCharacteristic">
    <xs:sequence>
      <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="heaviestAxleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
↪" maxOccurs="1" />
      <xs:element name="heaviestAxleWeightCharacteristicExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="HeightCharacteristic">
    <xs:sequence>
      <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="vehicleHeight" type="D2LogicalModel:MetresAsFloat" minOccurs=
↪"1" maxOccurs="1" />
      <xs:element name="heightCharacteristicExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="IndividualVehicleDataValues">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:TrafficData">
        <xs:sequence>
```

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```
          <xs:element name="individualVehicleSpeed" type="D2LogicalModel:SpeedValue" minOccurs="0" />
          <xs:element name="arrivalTime" type="D2LogicalModel:DateTimeValue" minOccurs="0" />
          <xs:element name="exitTime" type="D2LogicalModel:DateTimeValue" minOccurs="0" />
          <xs:element name="passageDurationTime" type="D2LogicalModel:DurationValue" minOccurs="0" />
          <xs:element name="presenceDurationTime" type="D2LogicalModel:DurationValue" minOccurs="0" />
          <xs:element name="timeGap" type="D2LogicalModel:DurationValue" minOccurs="0" />
          <xs:element name="timeHeadway" type="D2LogicalModel:DurationValue" minOccurs="0" />
          <xs:element name="distanceGap" type="D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
          <xs:element name="distanceHeadway" type="D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
          <xs:element name="individualVehicleDataValuesExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="InformationStatusEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="real" />
      <xs:enumeration value="securityExercise" />
      <xs:enumeration value="technicalExercise" />
      <xs:enumeration value="test" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="Integer">
    <xs:restriction base="xs:integer" />
  </xs:simpleType>
  <xs:complexType name="InternationalIdentifier">
    <xs:sequence>
      <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1" maxOccurs="1" />
      <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="KilometresPerHour">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:simpleType name="LaneEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="allLanesCompleteCarriageway" />
      <xs:enumeration value="busLane" />
      <xs:enumeration value="busStop" />
      <xs:enumeration value="carPoolLane" />
      <xs:enumeration value="centralReservation" />
      <xs:enumeration value="crawlerLane" />
```

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```
      <xs:enumeration value="emergencyLane" />
      <xs:enumeration value="escapeLane" />
      <xs:enumeration value="expressLane" />
      <xs:enumeration value="hardShoulder" />
      <xs:enumeration value="heavyVehicleLane" />
      <xs:enumeration value="lane1" />
      <xs:enumeration value="lane2" />
      <xs:enumeration value="lane3" />
      <xs:enumeration value="lane4" />
      <xs:enumeration value="lane5" />
      <xs:enumeration value="lane6" />
      <xs:enumeration value="lane7" />
      <xs:enumeration value="lane8" />
      <xs:enumeration value="lane9" />
      <xs:enumeration value="layBy" />
      <xs:enumeration value="leftHandTurningLane" />
      <xs:enumeration value="leftLane" />
      <xs:enumeration value="localTrafficLane" />
      <xs:enumeration value="middleLane" />
      <xs:enumeration value="opposingLanes" />
      <xs:enumeration value="overtakingLane" />
      <xs:enumeration value="rightHandTurningLane" />
      <xs:enumeration value="rightLane" />
      <xs:enumeration value="rushHourLane" />
      <xs:enumeration value="setDownArea" />
      <xs:enumeration value="slowVehicleLane" />
      <xs:enumeration value="throughTrafficLane" />
      <xs:enumeration value="tidalFlowLane" />
      <xs:enumeration value="turningLane" />
      <xs:enumeration value="verge" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="Language">
    <xs:restriction base="xs:language" />
  </xs:simpleType>
  <xs:complexType name="LengthCharacteristic">
    <xs:sequence>
      <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="vehicleLength" type="D2LogicalModel:MetresAsFloat" minOccurs=
↪"1" maxOccurs="1" />
      <xs:element name="lengthCharacteristicExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="LoadType2Enum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="refrigeratedGoods" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="LoadTypeEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="abnormalLoad" />
      <xs:enumeration value="ammunition" />
      <xs:enumeration value="chemicals" />
      <xs:enumeration value="combustibleMaterials" />
```

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```
      <xs:enumeration value="corrosiveMaterials" />
      <xs:enumeration value="debris" />
      <xs:enumeration value="empty" />
      <xs:enumeration value="explosiveMaterials" />
      <xs:enumeration value="extraHighLoad" />
      <xs:enumeration value="extraLongLoad" />
      <xs:enumeration value="extraWideLoad" />
      <xs:enumeration value="fuel" />
      <xs:enumeration value="glass" />
      <xs:enumeration value="goods" />
      <xs:enumeration value="hazardousMaterials" />
      <xs:enumeration value="liquid" />
      <xs:enumeration value="livestock" />
      <xs:enumeration value="materials" />
      <xs:enumeration value="materialsDangerousForPeople" />
      <xs:enumeration value="materialsDangerousForTheEnvironment" />
      <xs:enumeration value="materialsDangerousForWater" />
      <xs:enumeration value="oil" />
      <xs:enumeration value="ordinary" />
      <xs:enumeration value="perishableProducts" />
      <xs:enumeration value="petrol" />
      <xs:enumeration value="pharmaceuticalMaterials" />
      <xs:enumeration value="radioactiveMaterials" />
      <xs:enumeration value="refuse" />
      <xs:enumeration value="toxicMaterials" />
      <xs:enumeration value="vehicles" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:complexType name="LocationCharacteristicsOverride">
    <xs:sequence>
      <xs:element name="measurementLanesOverride" type="D2LogicalModel:LaneEnum" minOccurs="0" maxOccurs="1" />
      <xs:element name="reversedFlow" type="D2LogicalModel:Boolean" minOccurs="0" maxOccurs="1" />
      <xs:element name="locationCharacteristicsOverrideExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="MeasuredDataPublication">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:PayloadPublication">
        <xs:sequence>
          <xs:element name="measurementSiteTableReference" type="D2LogicalModel:_MeasurementSiteTableVersionedReference" minOccurs="1" maxOccurs="1" />
          <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
          <xs:element name="siteMeasurements" type="D2LogicalModel:SiteMeasurements" maxOccurs="unbounded" />
          <xs:element name="measuredDataPublicationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MeasuredValue">
```

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```
    <xs:sequence>
      <xs:element name="measurementEquipmentTypeUsed" type="D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
      <xs:element name="locationCharacteristicsOverride" type="D2LogicalModel:LocationCharacteristicsOverride" minOccurs="0" />
      <xs:element name="measurementEquipmentFault" type="D2LogicalModel:MeasurementEquipmentFault" minOccurs="0" maxOccurs="unbounded" />
      <xs:element name="basicData" type="D2LogicalModel:BasicData" minOccurs="0" />
      <xs:element name="measuredValueExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="MeasurementEquipmentFault">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:Fault">
        <xs:sequence>
          <xs:element name="measurementEquipmentFault" type="D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
          <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="MeasurementEquipmentFaultEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="intermittentDataValues" />
      <xs:enumeration value="noDataValuesAvailable" />
      <xs:enumeration value="spuriousUnreliableDataValues" />
      <xs:enumeration value="unspecifiedOrUnknownFault" />
      <xs:enumeration value="other" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="MetresAsFloat">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:complexType name="MultilingualString">
    <xs:sequence>
      <xs:element name="values">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="value" type="D2LogicalModel:MultilingualStringValue" maxOccurs="unbounded" />
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="MultilingualStringValue">
    <xs:simpleContent>
      <xs:extension base="D2LogicalModel:MultilingualStringValueType">
        <xs:attribute name="lang" type="xs:language" />
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
  <xs:simpleType name="MultilingualStringValueType">
```

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```
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="NonNegativeInteger">
    <xs:restriction base="xs:nonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="NumberOfAxlesCharacteristic">
    <xs:sequence>
      <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
      <xs:element name="numberOfAxles" type="D2LogicalModel:NonNegativeInteger"␣
↪minOccurs="1" maxOccurs="1" />
      <xs:element name="numberOfAxlesCharacteristicExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="OccupancyChangeValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
↪"1" maxOccurs="1" />
          <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="PassengerCarUnitsPerHour">
    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
  </xs:simpleType>
  <xs:complexType name="PayloadPublication" abstract="true">
    <xs:sequence>
      <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"␣
↪maxOccurs="1" />
      <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"␣
↪maxOccurs="1" />
      <xs:element name="publicationCreator" type=
↪"D2LogicalModel:InternationalIdentifier" />
      <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
    </xs:sequence>
    <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
  </xs:complexType>
  <xs:complexType name="PcuFlowValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="pcuFlowRate" type="D2LogicalModel:PassengerCarUnitsPerHour
↪" minOccurs="1" maxOccurs="1" />
          <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
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```
  </xs:complexType>
  <xs:simpleType name="Percentage">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:complexType name="PercentageValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="percentage" type="D2LogicalModel:Percentage" minOccurs="1
↪" maxOccurs="1" />
          <xs:element name="percentageValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="Seconds">
    <xs:restriction base="D2LogicalModel:Float" />
  </xs:simpleType>
  <xs:complexType name="SiteMeasurements">
    <xs:sequence>
      <xs:element name="measurementSiteReference" type="D2LogicalModel:_
↪MeasurementSiteRecordVersionedReference" minOccurs="1" maxOccurs="1" />
      <xs:element name="measurementTimeDefault" type="D2LogicalModel:DateTime"␣
↪minOccurs="1" maxOccurs="1" />
      <xs:element name="measuredValue" type="D2LogicalModel:_
↪SiteMeasurementsIndexMeasuredValue" minOccurs="0" maxOccurs="unbounded" />
      <xs:element name="siteMeasurementsExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="SpeedPercentile">
    <xs:sequence>
      <xs:element name="vehiclePercentage" type="D2LogicalModel:PercentageValue" />
      <xs:element name="speedPercentile" type="D2LogicalModel:SpeedValue" />
      <xs:element name="speedPercentileExtension" type="D2LogicalModel:_ExtensionType
↪" minOccurs="0" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="SpeedValue">
    <xs:complexContent>
      <xs:extension base="D2LogicalModel:DataValue">
        <xs:sequence>
          <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=
↪"1" maxOccurs="1" />
          <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:simpleType name="String">
    <xs:restriction base="xs:string">
      <xs:maxLength value="1024" />
    </xs:restriction>
  </xs:simpleType>
```

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```
  <xs:simpleType name="TimePrecisionEnum">
    <xs:restriction base="xs:string">
      <xs:enumeration value="tenthsOfSecond" />
      <xs:enumeration value="second" />
      <xs:enumeration value="minute" />
      <xs:enumeration value="quarterHour" />
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          <xs:element name="pcuFlow" type="D2LogicalModel:PcuFlowValue" minOccurs="0"␣
↪/>
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↪minOccurs="0" />
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      <xs:element name="heaviestAxleWeightCharacteristic" type="D2LogicalModel:HeaviestAxleWeightCharacteristic" minOccurs="0" maxOccurs="2" />
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      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
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      <xs:enumeration value="snowChainsInUse" />
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          <xs:element name="vmsFaultExtension" type="D2LogicalModel:_ExtensionType"␣
↪minOccurs="0" />
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↪ExtensionType" minOccurs="0" />
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</xs:schema>
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CHAPTER

SEVEN

CHANGELOG

version 1.0.1

- Textual revision of the documentation

version 1.0.0

- Initial format and documentation

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